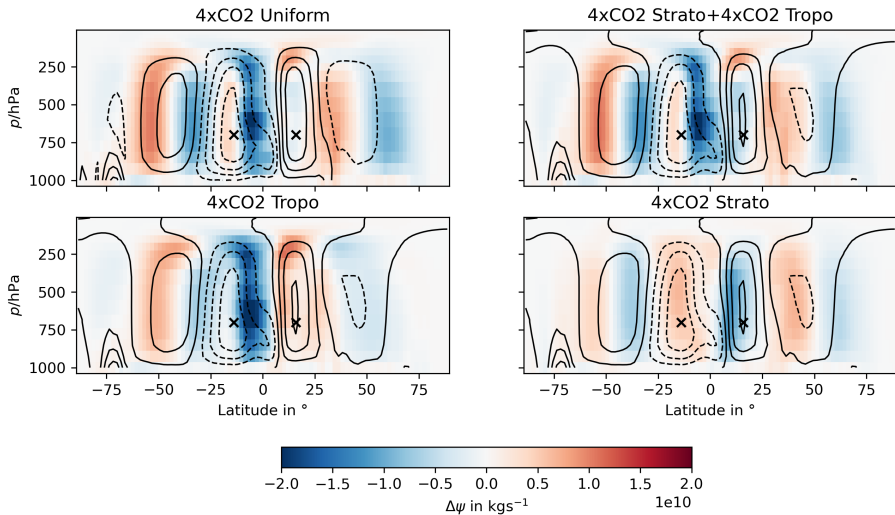


Ole Merkes

University of Hamburg

March 16, 2026

ECHAM mean meridional circulation



Changes in Hadley circulation

	PI Control	Uniform	Tropo	Strato
$\psi_{\max, \text{NH}}$	$7.72 \cdot 10^{10} \text{ kgs}^{-1}$	-1.6%	3.3%	-8.7%
$\psi_{\max, \text{SH}}$	$-9.64 \cdot 10^{10} \text{ kgs}^{-1}$	-1.4%	5.6%	-5.7%
$\Phi_{\text{H, NH}}$	29.2°	1.6°	0.76°	0.76°
$\Phi_{\text{H, SH}}$	-30.4°	-1.6°	-1.0°	-0.6°

Linearisation of H&H scaling

H&H

$$\Phi_H \sim \left(\frac{gH\Delta_h}{\Omega^2 R_E^2} \right)^{1/2} \quad (1)$$

$$\psi_{\max} \sim \frac{1}{\tau} \frac{g^{3/2} (H\Delta_h)^{5/2}}{\Omega^3 R_E^2 \Delta_v} \quad (2)$$

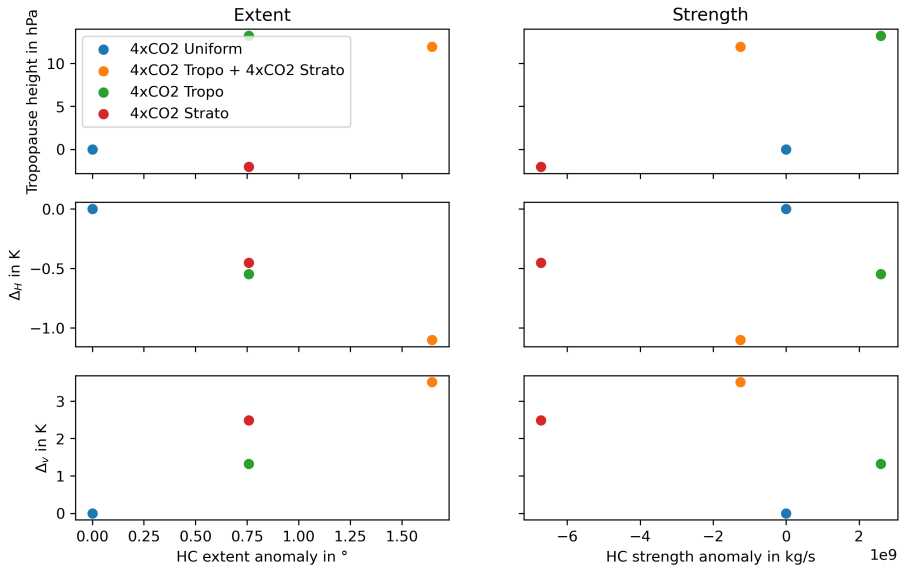
Linearisation

$$\frac{\delta\Phi_H}{\Phi_H} = \frac{1}{2} \frac{\delta H}{H} + \frac{1}{2} \frac{\delta\Delta_h}{\Delta_h} \quad (3)$$

$$\frac{\delta\psi}{\psi} = \frac{5}{2} \frac{\delta H}{H} + \frac{5}{2} \frac{\delta\Delta_h}{\Delta_h} - \frac{\delta\Delta_v}{\Delta_v} \quad (4)$$

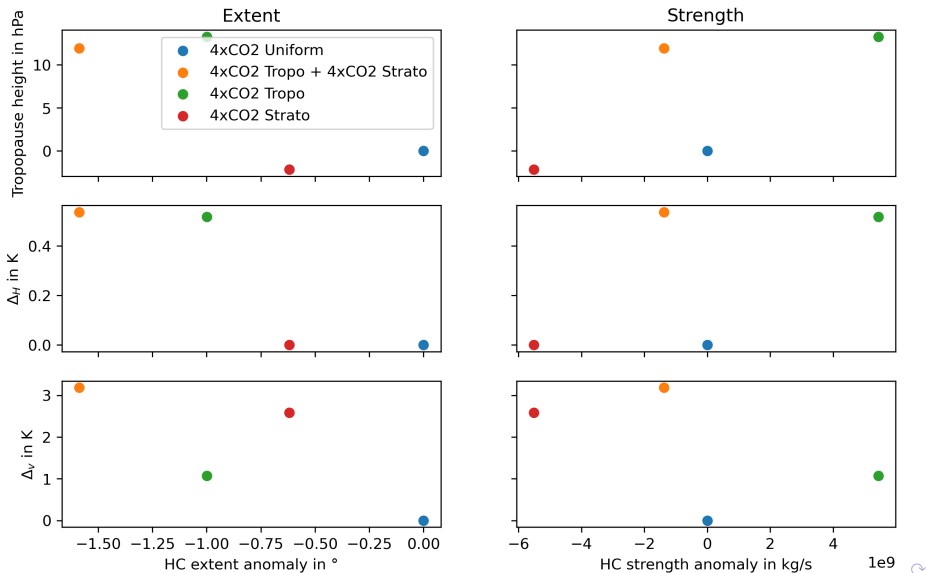
Scatter anomalies NH

Northern Hemisphere

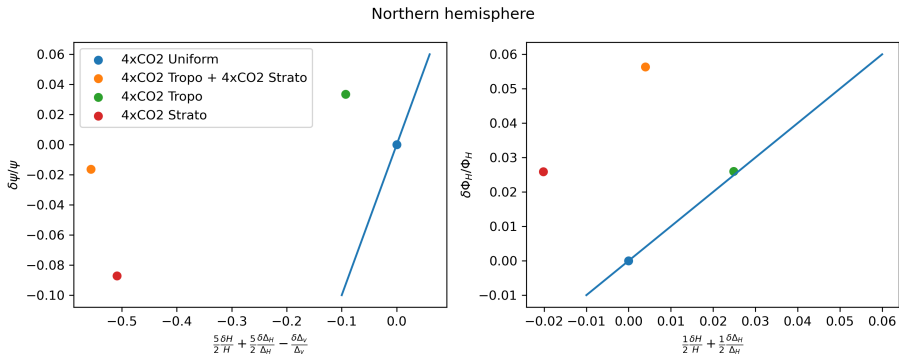


Scatter anomalies SH

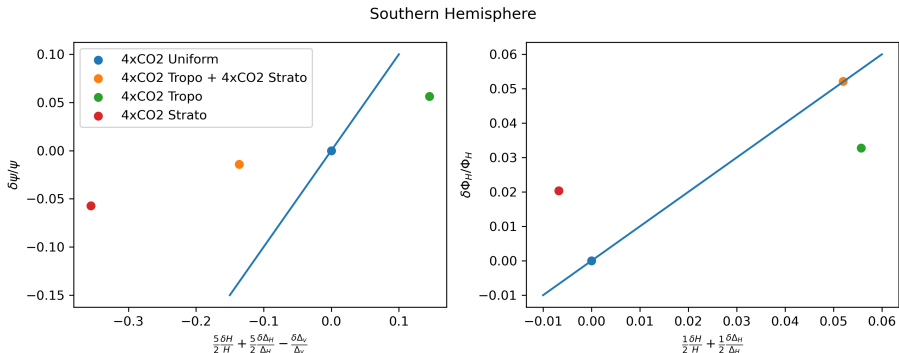
Southern Hemisphere



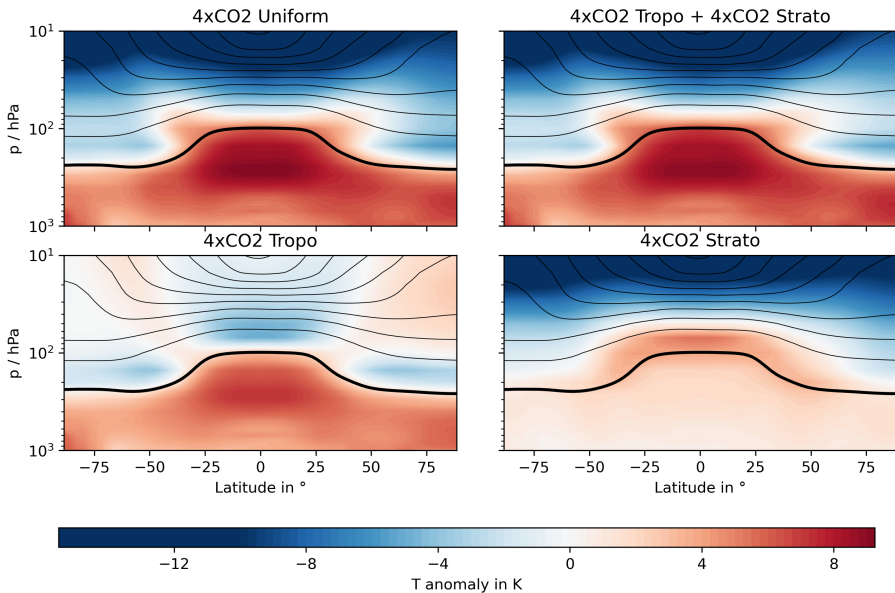
Applying H&H linearisation NH



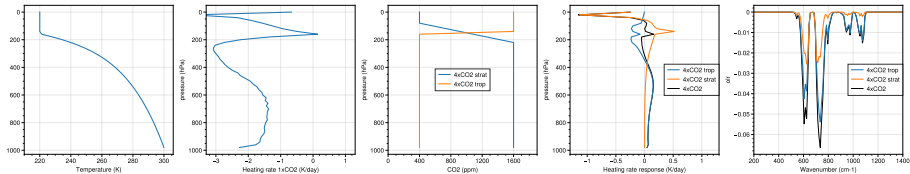
Applying H&H linearisation SH



Temperature anomalies



1D-RE simulation from Rodrigo Caballero



Heating rate

$$Q \sim \frac{\partial \tau}{\partial z} \quad (5)$$

Backup

Definitions D'Agostino et al. 2017

- meridional near-surface(averaged between the 700 and 800 hPa levels) potential temperature contrast
$$\Delta_h\theta(p, \varphi) = \bar{\theta}(800 - 700 \text{ hPa}, 0 - 10^\circ) - \bar{\theta}(800 - 700 \text{ hPa}, 40 - 60^\circ)$$
- subtropical near-surface static stability
$$\Delta_v\theta(p, \varphi) = \bar{\theta}(200 \text{ hPa}, 20 - 40^\circ) - \bar{\theta}(800 \text{ hPa}, 20 - 40^\circ)$$
- tropical tropopause pressure height $p_t(\varphi) = \bar{p}(15^\circ\text{S} - 15^\circ\text{N})$
- extratropical tropopause height $p_{et}(\varphi) = \bar{p}(35^\circ - 55^\circ)$ (Lu et al. 2007)